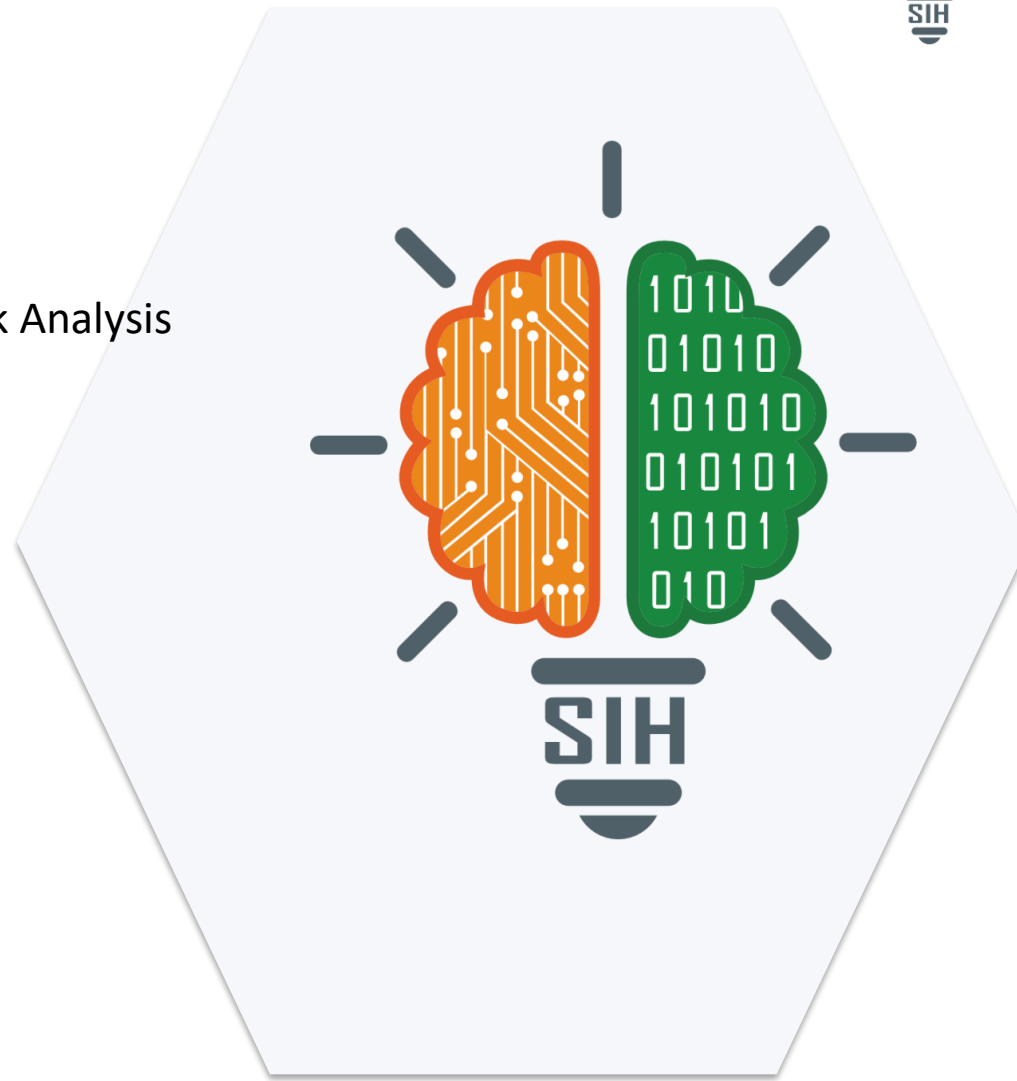


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – 26189**
- **Problem Statement Title - AI-Powered Criminal Network Analysis System**
- **Theme - Blockchain & Cybersecurity**
- **PS Category - Software**
- **Team ID - SIH2026-ST-842**
- **Team Name (Registered on portal) – Stryker**



IDEA / SOLUTION :

SENTINEL: AI-Powered Platform for Criminal Network Analysis

- **Combines Police Clues:** Automatically merges paper FIRs, phone call logs, and bank accounts into one unified visual map.
- **Catches Shadow Gang Bosses:** Graph AI algorithms uncover hidden leaders who order crimes without making direct phone calls.
- **Tracks Dirty Money:** Detects circular Hawala money loops and layered micro-transfers passed through dummy bank accounts.
- **Pinpoints Burner Phones:** Flags SIM card swapping in shared phone handsets and secret nocturnal suspect meetings (<500m).
- **CEO Prime Agent Swarm:** Autonomous AI commander directing specialized smart assistants to analyze cases 24/7.
- **Court-Certified Proof:** Seals evidence with SHA-256 blockchain hashing meeting 100% Section 63 BSA legal standards.

HOW IT SOLVES POLICE CHALLENGES :

❖ **Breaks Data Silos Across Stations :**

Police currently investigate FIRs, phone call logs, and bank statements in separate files. SENTINEL merges them into one 360° interactive suspect map in seconds.

❖ **Exposes Masterminds Behind Operatives :**

Crime kingpins rarely get their hands dirty or use known phones. By analyzing connection pathways, the AI pinpoints the hidden mastermind brokers.

❖ **Autonomous AI Investigation Swarm :**

Replaces slow manual paperwork. The CEO Prime Agent correlates clues across cases, calculates 0–100 danger scores, and alerts investigating officers instantly.

❖ **Guarantees Strong Court Convictions :**

Digital evidence often fails in court due to disputed tampering. SENTINEL generates tamper-proof PDF dossiers backed by blockchain verification for Section 63 BSA compliance.



Technical & Station Feasibility

- **Runs on Standard Station PCs:** Operates smoothly on normal police desktop computers without expensive cloud GPU clusters.
- **Direct Police Data Compatibility:** Natively ingests standard CCTNS FIR exports, telecom CDR spreadsheets, and bank statements.
- **Sub-Second Graph Query Speed:** Traverses complex networks of 10,000+ suspects in under 250 milliseconds (<0.25s).
- **Zero-Gas Blockchain Security:** Cryptographic SHA-256 hash chains guarantee court evidence integrity without cloud fees.

Economic Feasibility & Manpower Savings

- **100% Open-Source Foundation:** Built using FastAPI, NetworkX, and React 19—zero expensive recurring software licenses.
- **Compresses Weeks to Seconds:** Replaces weeks of manual paper cross-checking with instant automated intelligence.
- **Multiplies Police Productivity:** Allows lean investigative teams to crack large, multi-state organized crime syndicates.

Real-World Challenges & Practical Solutions

- **Messy & Dialect FIR Text:** Indian police FIRs contain spelling errors and local slang. Solved with custom Legal AI Named Entity Recognition + fuzzy matching (>88% accuracy).
- **Massive Telecom & Bank Volume:** Handling millions of call and transfer records without lag. Solved via indexed SQLite WAL and fast in-memory graph caching.

Benefits Across the Criminal Justice System

Law Enforcement & Frontline Police Officers :

- **1-Click Suspect Network Expansion:** Instantly reveals hidden gang associates, burner phones, and money trails.
- **Unmasks Shadow Kingpins:** Exposes gang bosses who orchestrate crimes without making direct phone calls.
- **Multi-Station Collaboration:** Breaks jurisdiction barriers by linking shared vehicles, weapons, and safehouses.

Crime Intelligence Analysts :

- **Autonomous Multi-Agent Swarm:** CEO Prime Agent automates tedious cross-referencing across FIRs and telecom logs.
- **Automated 0-100 Threat Scoring:** Prioritizes violent offenders and automatically detects Hawala laundering cycles.

Judiciary & Legal Prosecution :

- **100% Section 63 BSA Compliance:** Cryptographic SHA-256 blockchain proof guarantees court admissibility.
- **Rapid Charge-Sheet Preparation:** Generates certified forensic PDF dossiers that accelerate trials and eliminate delays.

"SENTINEL transforms fragmented police records into real-time operational clarity—unmasking shadow kingpins, stopping dirty money trails, and delivering faster, bulletproof convictions."

KEY OPERATIONAL OUTCOMES

- ⚡ **90% Time Reduction :** From weeks of paper cross-checking to seconds of graph analytics.
- 🔒 **100% Chain-of-Custody :** Cryptographically sealed under Section 63 BSA for undisputed trials.
- 🔍 **Zero Kingpin Blindspots :** Algorithmic unmasking of shadow brokers coordinating street operatives.



Academic, Legal & Operational Frameworks

1. Network Science & Covert Syndicate Analysis :

Betweenness Centrality (Freeman, 1977) & PageRank algorithms optimized for identifying high-influence brokers and shadow leaders in criminal networks.

2. Multi-Agent Autonomous Intelligence :

Agency Swarm orchestration architecture dividing investigative labor among specialized autonomous agents (Data Intel, Network Analysis, Threat Scoring, Forensic Reporting).

3. Indian Statutory Forensic Standards :

Section 63 of Bharatiya Sakshya Adhiniyam (BSA, 2023) / Section 65B Indian Evidence Act compliant digital hash chains ensuring tamper-evident chain of custody.

4. Financial Crime & Hawala Detection :

Financial Action Task Force (FATF) and Reserve Bank of India (RBI) typologies for detecting circular layering loops and smurfing transactions (<₹50,000 threshold).

⚡ **Working Prototype Status:** Live full-stack FastAPI + React 19 platform pre-loaded with 372 suspect nodes and 869 evidence links. Ready for live demonstration.

THANK YOU • Team Stryker • Problem Statement ID: 26189 • Open for Live Demonstration & Q&A